

Research project Bedrock & MCP Server

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Agenda

1. Project idea - Barcelona Smart City
2. Amazon Bedrock
3. MCP Server
4. Live Demonstration

Project idea - Barcelona Smart City

Main goal: Make data/information about Barcelona more accessible

Solution: Use AI chatbots to query real time and historical data to present information

Datasources:

- Weather
- Transportation
- Air quality
- Noise



Open Data BCN

Ajuntament de Barcelona's open data service

Amazon Bedrock

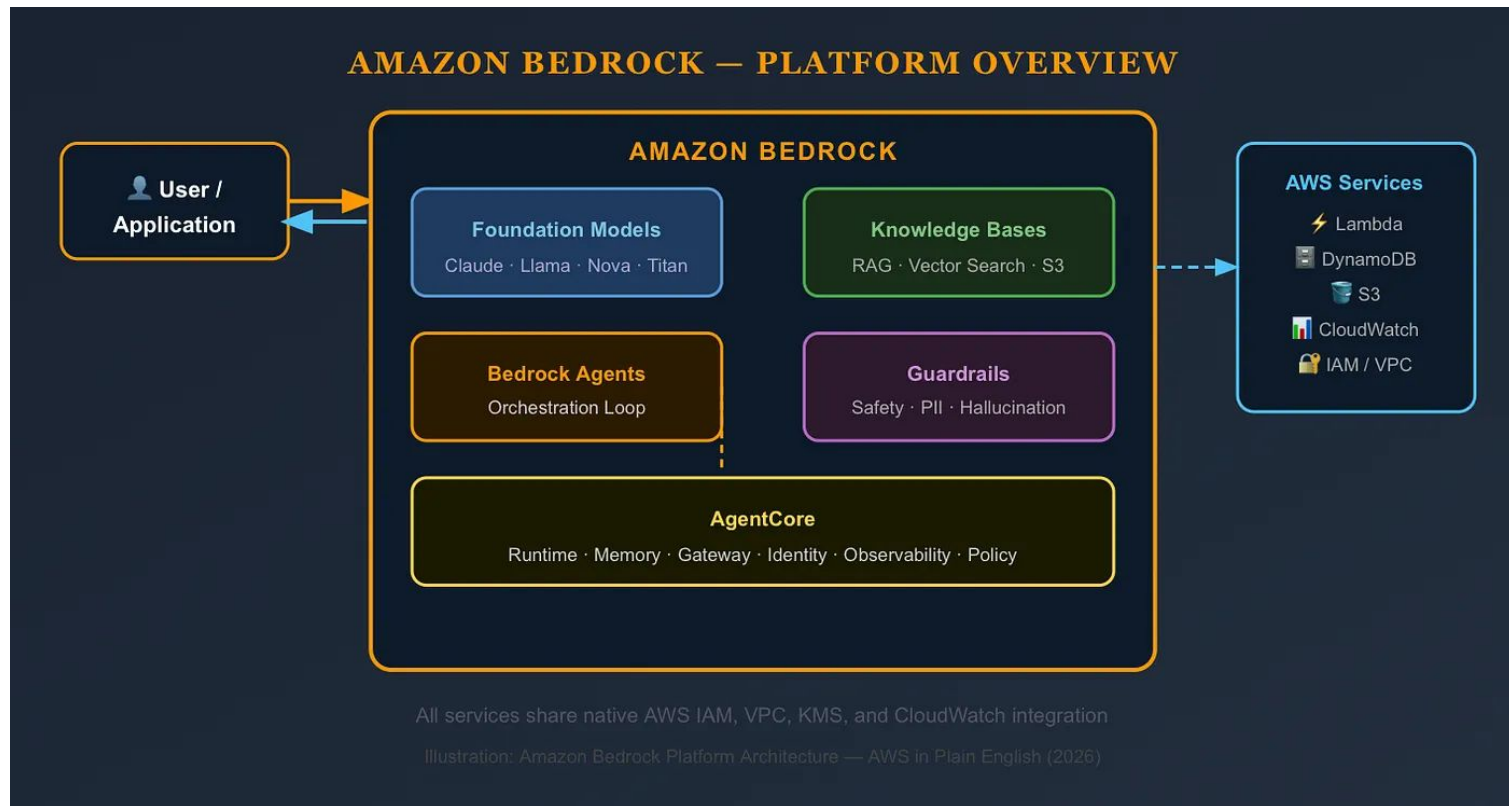
Core idea:

- Access foundation models through API
- Build AI apps securely inside the AWS ecosystem
- Avoid training, model hosting, or scaling

Benefits:

- Simple Management
- Fast Time to Market
- Scalability
- Security and Compliance
- Integration with AWS Services
- Cost Efficient

Amazon Bedrock



Model Context Protocol Server

Core idea:

- One server works with every AI client
- MCP = "USB-C for AI tools"

Three Roles:

- Host (AI app)
- Client (manages one server connection)
- MCP Server (Code)

Three primitives:

- Tools (functions the LLM can call)
- Resources (read-only data)
- Prompts (reusable prompt templates)

Model Context Protocol Server

Protocol: JSON-RPC 2.0

Step 1 – Handshake:

Client sends "initialize" with protocol version

Server replies with capabilities and server info

Step 2 – Tool discovery:

Client calls "tools/list"

Server returns name, description, inputSchema

Step 3 – Tool execution:

Client calls "tools/call" with arguments

Server executes and returns result as JSON

LLM incorporates result into its response